SAJAL AGARWAL

Senior Data Scientist | Generative AI Engineer

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PROFESSIONAL SUMMARY

Senior Data Scientist and Generative AI Engineer with 15+ years across machine learning, statistical modeling, and cloud data engineering. Builds and ships **production** systems: a cloud cost anomaly-detection platform that delivered **$12.8M CAD in annual savings** at Manulife, and a **grounded multi-agent RAG service** running on AKS that answers enterprise questions with citations and a confidence gate — and refuses when evidence is weak rather than hallucinating. Equally comfortable in regulated modeling (credit risk scorecards, fraud detection, insurance pricing) and in LLM system design (hybrid FAISS + BM25 retrieval, reranking, evaluation, cost attribution). Strong track record translating model output into decisions for finance and C-suite stakeholders.

TECHNICAL SKILLS

GenAI / LLM Systems	RAG pipelines, Hybrid retrieval (FAISS + BM25), Reciprocal Rank Fusion, Reranking, Agentic systems & tool calling, LangChain, Azure AI Foundry, Azure OpenAI / GPT-4, HuggingFace, Prompt engineering, LLM evaluation & golden evals
ML / Statistical Modeling	Scikit-learn, XGBoost, Random Forest, Facebook Prophet, SMOTE, SHAP, Time-series forecasting, Classification, Regression, Clustering, PCA, A/B testing
Cloud Platform	Microsoft Azure — Data Factory, Databricks, ADLS Gen2, Azure OpenAI / AI Foundry, Azure Functions, Azure Kubernetes Service (AKS)
Data Engineering	PySpark, Delta Lake, ETL/ELT pipeline design, Dimensional & snapshot modeling, Pandas, NumPy
Databases	PostgreSQL, MS SQL Server, MySQL, SQLite, IBM Cognos TM1
Reporting & BI	Power BI (DAX, Power Query, semantic models, incremental refresh), Grafana, Tableau, Qlik Sense, Advanced Excel, SSRS
Engineering & DevOps	Python, Flask, FastAPI, Streamlit, Docker, Kubernetes, Istio, REST APIs, Git, CI/CD, New Relic
Languages	Python, SQL, R
Domains	BFSI, Insurance, Healthcare, Cloud FinOps, Retail, CPG/FMCG, Aviation, Manufacturing

PROFESSIONAL EXPERIENCE

Data Scientist | Client: Manulife

Tata Consultancy Services — Associate Consultant, Analytics & Insights *Jul 2020 – Present | Toronto, ON*

- **Delivered $12.8M CAD in annual savings** by designing and deploying a production ML-based anomaly detection system that forecast expected Azure spend per cost centre and alerted owners to deviations with resource-level breakdowns.
- Designed and deployed a **production multi-agent RAG service** (Flask, FAISS + BM25, Azure AI Foundry) on **AKS behind Istio**, engineered so answers are citation-backed and the system refuses below a confidence threshold rather than producing unverifiable output.
- Built an enterprise **FinOps chargeback platform** on **Databricks / PySpark, ADF and ADLS Gen2**, delivering auditable cost-centre attribution, fair-share allocation of shared savings, and invoice reconciliation.
- Designed and deployed **end-to-end ML pipelines** for credit risk, insurance pricing, fraud detection and cloud cost optimization, owning the full lifecycle: development, validation, deployment and monitoring.
- Built **BI and observability dashboards** in Grafana and Power BI over complex PostgreSQL models, including snapshot-aware data-quality reporting and query-performance tracking across 50+ dashboards.
- Automated Azure billing and reporting with **Python and Azure Functions**; implemented ETL pipelines in **Azure Data Factory** for scheduled refreshes.
- Presented model outputs, cost drivers and remediation recommendations to senior leadership and finance stakeholders.

Production Systems & Platforms (Manulife)

1. ONE API Agent — Grounded Multi-Agent Service (Flask, FAISS + BM25, Azure AI Foundry, AKS)

- Built a **multi-agent service** answering natural-language questions across four enterprise domains (application inventory, cost centres, platform documentation, incidents) behind a single **OpenAI Assistants-compatible protocol** with Entra ID / MSAL authentication.
- Engineered a **zero-hallucination retrieval pipeline**: hybrid BM25 + FAISS retrieval fused with **Reciprocal Rank Fusion**, per-chunk score filtering, a transparent four-factor confidence gate, tool-less constrained generation, and post-generation **citation enforcement** that discards answers whose citations cannot be verified.
- Deployed on **AKS with Istio**; instrumented every query (answered or refused) to New Relic and guarded against regressions with **golden evaluation tests in CI**.
- GitHub: github.com/Sajalagarwal-ca/one-api-agent

2. Cloud Cost Anomaly Detection & Optimization (Azure, Facebook Prophet) — Flagship

- Designed and deployed an **ML anomaly detection system** identifying abnormal Azure spend at cost-centre level, delivering **$12.8M CAD in annual savings**.
- Built daily **Facebook Prophet** forecasts per cost centre with prediction-interval-based alerting, chosen for explainable trend/seasonality decomposition that finance could defend.
- Automated owner-targeted email alerts with resource-level variance breakdowns, converting detection into remediation.
- GitHub: github.com/Sajalagarwal-ca/Time-Series-Forecasting-via-meta-prophet-model

3. Azure FinOps & Chargeback Platform (Databricks, PySpark, ADF, ADLS Gen2, PostgreSQL)

- Built an enterprise chargeback platform delivering **auditable cost-centre attribution**, fair-share allocation of shared savings and discounts, invoice reconciliation, and daily showback to budget owners.
- Implemented cost-centre resolution as an **ordered waterfall** (finance override → validated tag → inventory/parent mapping → history → regional default), recording the resolving rule on every record so any disputed allocation is traceable.

4. Grafana AI Cost Observability (Grafana, PostgreSQL, Azure Data Factory, Python)

- Unified **Gen AI, GitHub Copilot and Copilot Studio** spend — three incompatible billing models (consumption meters, per-seat licences, consumed credits) — into one cost-centre-level dashboard.
- Normalised currency, converted credits to charges and resolved every record to a cost centre in a tested Python transformation layer, keeping the dashboard a thin read over reconciled tables.

5. Application Data Quality Reporting (Power BI, Power Query, DAX, SQL)

- Modelled daily application-inventory snapshots into a **snapshot-aware** completeness and business-rule compliance report, reworking grain and date-aware relationships so historical comparisons do not double-count.
- Complemented by a **query-performance tracking** solution across 50+ Grafana dashboards using PostgreSQL statistics, ranking optimisation targets by total accumulated time rather than worst-case duration.

Applied Machine Learning Projects

1. Credit Risk Modeling - Probability of Default Scorecard (Logistic Regression, Streamlit)

- Built a **probability-of-default scorecard** for small-finance lending, where borrowers often lack formal credit-bureau history; engineered behavioural ratios including **loan-to-income, delinquency ratio and average days-past-due per delinquency**.
- Applied credit-industry feature selection - **VIF** for multicollinearity and **Weight of Evidence / Information Value** for categorical predictors; validated on **ROC-AUC, Gini, KS and decile rank-ordering**.
- Benchmarked **XGBoost as a challenger** with Optuna tuning and selected logistic regression for production, since readable coefficients are required for adverse-action reasoning in regulated lending.
- Deployed as a **Streamlit application** returning a default probability, a 300-900 credit score and a risk rating band.
- GitHub: github.com/Sajalagarwal-ca/credit_risk_modeling | Live Demo: creditriskmodeling-dbvgyaavxuqjfbjtztxep5.streamlit.app

2. Fraudulent Loan Application Detection (Random Forest, SMOTE)

- Built a fraud classifier on severely imbalanced loan data, applying **SMOTE within cross-validation folds** to avoid leakage and tuning via GridSearchCV.

- Achieved **F1 above 0.75**, evaluated on precision-recall rather than accuracy; identified FICO score, debt-to-income ratio and loan purpose as dominant indicators.
- GitHub: github.com/Sajalagarwal-ca/Fraudulent-Loan-Application-Detection-Project

3. Health Insurance Premium Calculator (XGBoost, scikit-learn, SHAP, Streamlit)

- Predicted premiums from age, medical history, lifestyle and demographics; evaluated with RMSE / MAE / R^2 and **SHAP per-prediction attribution** so individual quotes are explainable.
- GitHub: github.com/Sajalagarwal-ca/healthcare_premium_prediction | Live Demo: healthcarepremiumprediction-ufs9ahmsfgntfmn6lqrkvf.streamlit.app

Generative AI Projects

1. Natural Language to SQL Query Engine (LangChain, PostgreSQL, Streamlit)

- Built a natural-language interface over PostgreSQL using LangChain's SQL agent with schema introspection and conversational memory; generated SQL always surfaced to the user, executed through a read-only credential.
- GitHub: github.com/Sajalagarwal-ca/langchain_query_NL_PostgreSQL_DB

2. Personal RAG Document Assistant (Streamlit, OpenAI, FAISS, HuggingFace)

- Local RAG pipeline over multi-format documents (PDF, Word, Excel, CSV, TXT) with all-MiniLM-L6-v2 embeddings and FAISS, returning grounded answers with source file, page, chunk ID and snippet.
- GitHub: github.com/Sajalagarwal-ca/RAG-Document-reader

3. Insurance Fraud Investigation Assistant (Azure OpenAI GPT-4, LangChain, FAISS, Power BI)

- Summarised long claim documents and surfaced anomalies against historical fraud patterns, combining structured claims data with unstructured emails and scanned PDFs; anomalies visualised in Power BI for audit-ready investigation summaries.

4. Additional applied AI projects

- **RAG Telecom Support Agent** (LangChain, Groq, ChromaDB) — domain support chatbot over FAQs, resolved tickets and user guides with source-type filtering. | **AI Shopping Agent** (Streamlit, LangChain, Groq) — multimodal product discovery via text or image with explicit order confirmation. | **AI Travel Assistant** (LangChain, Azure OpenAI, Google Places API) — itinerary planning grounded on live API data.

Data Scientist | Client: Tata Steel

Tata Consultancy Services — Assistant Consultant, Analytics & Insights *Jun 2019 – Jul 2020 | Kolkata, India*

- Applied statistical models to optimise the mild steel door production process, improving yield.
- Developed supervised and unsupervised models (regression, clustering, PCA) to diagnose process deviations.
- Designed forecasting and classification models using multivariate regression, logistic regression, decision trees and time-series analysis.
- Experimented with model pruning and quantization to optimise predictive maintenance models for edge deployment.
- Built and maintained KPI dashboards for cross-functional teams.

Manager — Analytics

GMR — Delhi International Airport Ltd. *Jul 2018 – Jun 2019 | New Delhi, India*

- Consolidated management reporting across all airport terminals for Non-Aero Commercial reporting to CCO/CEO on Qlik Sense and Power BI.
- Designed data-driven campaigns generating **INR 2 Million in incremental revenue** across retail and F&B segments.

Deputy Manager — Analytics

WNS Global Services *Dec 2017 – Jul 2018 | Gurugram, India*

- Partnered with UK and USA clients across Healthcare and FMCG analytics verticals.
- Led a team of 5 analysts for financial and commercial reporting on IMS/Nielsen/IRI data in Power BI.
- Conducted sales and competitive landscape analysis identifying high-growth consumer healthcare segments across SE Asia.
- Tracked and reported KPIs (Gross Sales, COGS, Net Sales, Operating Profit, Net Profit) from public financial data.

EDUCATION

Qualification	Institution	Year	Grade
Bachelor of Engineering	Jaypee Institute of Information Technology, Noida	2009	6.8 / 10
MBA	Jaypee Business School	2010	7.3 / 10
Certificate – R Programming	Indian Institute of Technology, Chennai	2017	75%
Diploma in Statistics	Indira Gandhi National Open University	2018	73%

CERTIFICATIONS

- Microsoft Certified: Power BI Data Analyst Associate — Microsoft (2023)
- Data Science Using Python — Udemy (2021)
- Python for Data Science — Udemy (2020)